



UNIVERSITY OF CAPE TOWN

CSC2042S: Machine Learning

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1 Supervised learning

1.1 What is machine learning?

Definition: Machine Learning

Machine learning is the study of algorithms that:

- improve their performance P (according to a criterion that measures success)
- at some task T
- with experience E (from data examples).

A well-defined task is given by $\langle P, T, E \rangle$. For example:

- **Image classification:** $\langle \text{Accuracy, Classify handwritten digits, Dataset of labelled images} \rangle$.
- **Chess:** $\langle \text{Playing quality, Play chess against human, Dataset of old games} \rangle$.
- **ChatGPT:** $\langle \text{Quality of conversation, Generate responses, All text on the web} \rangle$.

1.2 Types of machine learning

There are many types:

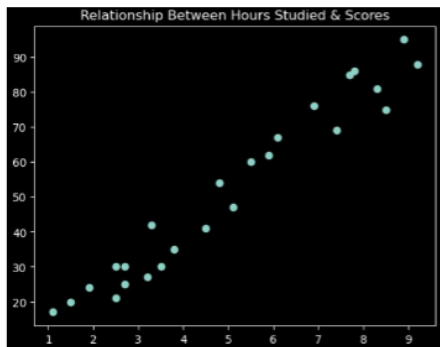
- Supervised learning.
- Unsupervised learning.
- Reinforcement learning.
- etc.

1.2.1 Unsupervised learning

The problem is framed as finding structure in x without labels (no y). Inputs are examples in the dataset represented by *features*. Data D is $|D|$ examples $\{x_1, x_2, \dots, x_{|D|}\}$ without any labels. Given a new example x , where does this fit into the data?

1.3 Simple linear regression

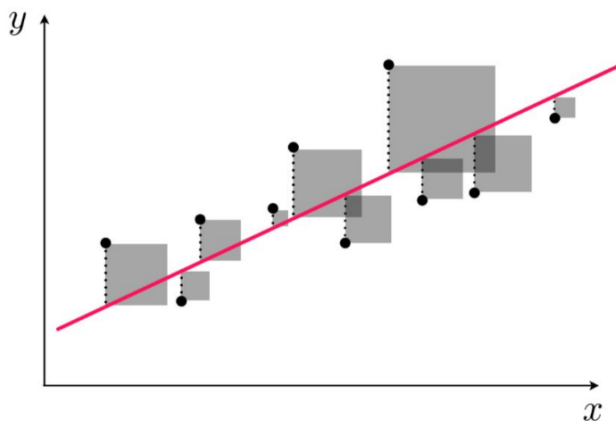
A simple linear regression model predicts the output as a linear function of one input feature x : $y = wx + b$, otherwise known as a straight line function. Changes in x imply a proportional change in y . Even if the relationship is not perfectly linear, linear regression can still be effective as an approximation or baseline model.



1.3.1 Loss function

For each input, linear regression produces an estimate $\hat{y}_i = w \cdot x_i + b$. We want to set w and b to minimize the distance between our estimate $\hat{y}_i$ (prediction) and the true y_i . We want to minimize this distance/error across our entire training set $\{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$.

A principled way to define the learning problem is with a loss function (also known as a cost function): $L(\hat{y}_i, y_i)$ measuring how close the model prediction $\hat{y}_i$ is to the observed data y_i . If the model fits the data well, so that $y_i \approx \hat{y}_i$, then the loss function should take a small value, and vice versa. For this, we can use the squared error loss.



The squared error loss measures how good your linear regression model with parameters w, b fits the data. For a single data point (x_i, y_i) :

$$\begin{aligned} L(\hat{y}_i, y_i) &= (y_i - \hat{y}_i)^2 \\ &= (y_i - f(x_i; w, b))^2 \\ &= (y_i - (w \cdot x + b))^2 \\ &= (y_i - w \cdot x - b)^2 \end{aligned}$$

We want to find the weight that minimizes the loss, averaged over all examples in our training set.

$$J(w, b) = \frac{1}{N} \sum_{i=1}^N L(\hat{y}_i, y_i)$$

Note: Why as a function of w and b ?

These are the variables we want to learn!

All examples (x_i, y_i) in the function are determined by whatever is in the training set. This is out of our control. Now, we want to find:

$$\hat{w}, \hat{b} = \underset{w, b}{\operatorname{argmin}} J(w, b)$$

For linear regression, the loss function (mean squared error) is convex. This means it has a single global minimum and no local minima. Linear regression has a closed form solution. The question is, how do we find $\hat{w}, \hat{b} = \underset{w, b}{\operatorname{argmin}} J(w, b)$?

1. Take the partial derivative of J with respect to w and b .
2. Set the partial derivative equal to 0.
3. Solve for w and b .

Derivation

Derivation w.r.t b :

$$\begin{aligned}\frac{\partial}{\partial b} J(w, b) &= \frac{\partial}{\partial b} \frac{1}{N} \sum_{i=1}^N (y_i - wx_i - b)^2 \\ &= \frac{-2}{N} \sum_{i=1}^N (y_i - wx_i - b) \\ 0 &= \sum_{i=1}^N (y_i - wx_i - b) \quad (\text{Setting equal to 0}) \\ 0 &= \sum_{i=1}^N y_i - w \sum_{i=1}^N x_i - Nb \\ \hat{b} &= \bar{y} - \hat{w}\bar{x}\end{aligned}$$

Derivation w.r.t w :

Before differentiating w.r.t w , substitute this expression for b back into the cost so that J becomes a function of w only:

$$J(w) = \frac{1}{N} \sum_{i=1}^N \left[y_i - wx_i - (\bar{y} - w\bar{x}) \right]^2 = \frac{1}{N} \sum_{i=1}^N \left[(y_i - \bar{y}) - w(x_i - \bar{x}) \right]^2$$

Now differentiate w.r.t w :

$$\frac{d}{dw} J(w) = \frac{-2}{N} \sum_{i=1}^N \left[(y_i - \bar{y}) - w(x_i - \bar{x}) \right] (x_i - \bar{x})$$

Set the derivative to 0 and simplify:

$$0 = \frac{-2}{N} \left[\sum_{i=1}^N (y_i - \bar{y})(x_i - \bar{x}) - w \sum_{i=1}^N (x_i - \bar{x})^2 \right]$$

$$w \sum_{i=1}^N (x_i - \bar{x})^2 = \sum_{i=1}^N (y_i - \bar{y})(x_i - \bar{x})$$

$$\hat{w} = \frac{\sum_{i=1}^N (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

1.4 Logistic regression

1.4.1 Probability basics

- $P(A)$: probability of event A .
- $P(A^c)$: probability of event A not happening.
- $P(A|B)$: probability of A given B (conditional probability).
- For any event A , $0 \leq P(A) \leq 1$.
- Let S be the sample space of all possible events, then $P(S) = 1$.
- For any event A , $P(A^c) = 1 - P(A) \iff P(A) = 1 - P(A^c)$.

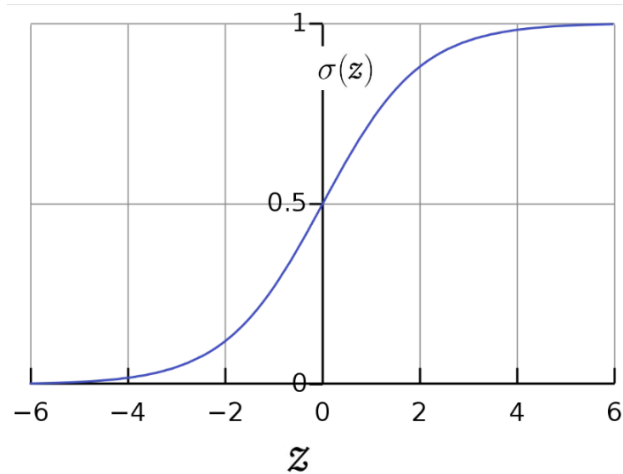
1.4.2 Discriminative classifier

Discriminative models learn boundaries between classes. For example, suppose we are classifying cat vs. dog images. The input x is the image (matrix of pixel values). The output y is the class (either cat or dog). The goal of classification is to find the correct class y given input x . To classify, we compute: $y = \operatorname{argmax}_{y \in C} P(y|x)$.

1.4.3 Logistic regression

We introduce the sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



The sigmoid function is called a logistic function. It squeezes any input value between 0 and 1. To classify the input $\mathbf{x}$:

- Compute the weighted sum, $z = \sum w_i x_i + b = \mathbf{w} \cdot \mathbf{x} + b$.
- Pass z through the sigmoid function, $\sigma(\mathbf{w} \cdot \mathbf{x} + b)$.
- Treat the result as a probability.

$$P(y = 1|\mathbf{x}) = \sigma(\mathbf{w} \cdot \mathbf{x} + b) = \frac{1}{1 + \exp(-\mathbf{w} \cdot \mathbf{x} - b)}$$

A key property of the sigmoid function: $1 - \sigma(z) = \sigma(-z)$.

Derivation

$$\begin{aligned} P(y = 0|\mathbf{x}) &= 1 - \sigma(\mathbf{w} \cdot \mathbf{x} + b) \\ &= 1 - \frac{1}{1 + \exp(-\mathbf{w} \cdot \mathbf{x} - b)} \\ &= \frac{\exp(-\mathbf{w} \cdot \mathbf{x} - b)}{1 + \exp(-\mathbf{w} \cdot \mathbf{x} - b)} \times \frac{\exp(\mathbf{w} \cdot \mathbf{x} + b)}{\exp(\mathbf{w} \cdot \mathbf{x} + b)} \\ &= \frac{1}{1 + \exp(\mathbf{w} \cdot \mathbf{x} + b)} \\ &= \sigma(-\mathbf{w} \cdot \mathbf{x} - b) \end{aligned}$$

Now, we have a way to compute both $P(y = 1|\mathbf{x})$ and $P(y = 0|\mathbf{x})$. We can now make decisions:

$$\text{decision}(\mathbf{x}) = \begin{cases} 1 & \text{if } P(y = 1|\mathbf{x}) > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

The decision boundary of $P(y|x) = 0.5$ is defined where $\mathbf{w} \cdot \mathbf{x} + b = 0$. This is the default decision boundary in logistic regression. However, you can specify a different decision boundary to suit your task.

- Choose a higher value, like $P(y = 1|\mathbf{x}) > 0.9$, if you want the model to be confident before flagging an instance of a positive class.
- Choose a lower value, like $P(y = 1|\mathbf{x}) > 0.2$, if you don't want to risk missing instances of the positive class (e.g. medical diagnosis).

1.4.4 Example

Predict if a fire weather warning should be announced.

- x_1 : temperature in C° .
- x_2 : wind speed in km/h.
- $y = 0$: no warning needed.
- $y = 1$: warning needed.

Suppose the weights are $\mathbf{w} = [0.1, 0.05]$ and the bias term is $b = -5$. For an input vector $\mathbf{x} = [25, 60]$,

$$P(y = 1|\mathbf{x}) = \sigma(\mathbf{w} \cdot \mathbf{x} + b) = \sigma((0.1)(25) + (0.05)(60) - 5) = \sigma(0.5) = 0.622$$

$$P(y = 0|\mathbf{x}) = 1 - P(y = 1|\mathbf{x}) = 1 - 0.622 = 0.378$$

Since $P(y = 1|\mathbf{x}) = 0.622 > 0.5$, a warning should be given.

1.5 Cross-entropy loss

1.5.1 Log probabilities

Definition: Log probability

A log probability is a logarithm of a probability value, $\log P(X)$. This represents probabilities on a logarithmic scale $(-\infty, 0]$ instead of the usual $[0, 1]$ probability interval.

A useful property of log properties:

$$\log P(ABC) = \log(P(A) \times P(B) \times P(C)) = \log P(A) + \log P(B) + \log P(C)$$

1.5.2 Loss function

We use conditional **maximum likelihood estimation**:

- Find parameters that **maximize the probability of the true labels** in the training data given the observations.
- For binary classification, since there are 2 discrete outcomes, we can express the classifier's probability of the true label y as:

$$P(y|\mathbf{x}) = \hat{y}^y (1 - \hat{y})^{1-y}$$

- If $y = 1$, this simplifies to $\hat{y}$ which is $P(y = 1|\mathbf{x})$.
- If $y = 0$, this simplifies to $1 - \hat{y}$ which is $P(y = 0|\mathbf{x})$.

We can rewrite this in terms of log probabilities:

$$\begin{aligned}\log P(y|\mathbf{x}) &= \log [\hat{y}^y (1 - \hat{y}^{1-y})] \\ &= y \log \hat{y} + (1 - y) \log(1 - \hat{y})\end{aligned}$$

We can then turn this into a loss function to be minimized by using the negative log likelihood. This is called the **cross-entropy loss** between the prediction and the true class.

$$\begin{aligned}L_{\text{CE}}(\hat{y}, y) &= -\log P(y|\mathbf{x}) \\ &= -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})] \\ &= -[y \log[\sigma(\mathbf{w} \cdot \mathbf{x} + b)] + (1 - y) \log[\sigma(-\mathbf{w} \cdot \mathbf{x} - b)]]\end{aligned}$$

By minimizing this loss, we maximize the probabilities assigned to the true labels by the logistic regression model.

1.5.3 Example

Predict if a fire weather warning should be announced.

- x_1 : temperature in C° .
- x_2 : wind speed in km/h.
- $y = 0$: no warning needed.
- $y = 1$: warning needed.

Suppose the weights are $\mathbf{w} = [0.1, 0.05]$ and the bias term is $b = -5$.

For an input vector $\mathbf{x} = [25, 60]$ and true label $y = 1$:

$$\begin{aligned}P(y = 1|\mathbf{x}) &= \sigma(\mathbf{w} \cdot \mathbf{x} + b) = 0.622 \\ L_{\text{CE}}(\hat{y}, y) &= -\log P(y = 1|\mathbf{x}) = -\log 0.622 = 0.206\end{aligned}$$

For an input vector $\mathbf{x} = [25, 60]$ and true label $y = 0$:

$$\begin{aligned}P(y = 0|\mathbf{x}) &= 1 - \sigma(\mathbf{w} \cdot \mathbf{x} + b) = 0.378 \\ L_{\text{CE}}(\hat{y}, y) &= -\log P(y = 0|\mathbf{x}) = -\log 0.378 = 0.423\end{aligned}$$

Since the loss of true label $y = 1$ is lower, we should give a warning.

Note

The loss is higher if the model is wrong. Lower probabilities assigned to the true class implies a higher loss.

1.5.4 Objective function

Given the cross-entropy loss, we want to minimize this loss on average across our training set:

$$\frac{1}{N} \sum_{i=1}^N L_{\text{CE}}(\hat{y}_i, y_i)$$

If we can adjust parameters $\mathbf{w}$ and b to lower the loss over all instances in the training set, we will have a good classifier. This is the essence of modern machine learning and deep learning.

1.5.5 Optimization

Let the loss function be parameterized by parameters $\theta = (\mathbf{w}, b)$. We represent $\hat{y}$ as $f(\mathbf{x}; \theta)$. We want to find parameters that minimize the loss averaged over all examples in our training set.

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \frac{1}{N} \sum_{i=1}^N L_{\text{CE}}(f(\mathbf{x}; \theta), y_i)$$

There is **no closed form solution** for finding minimizing weights for logistic regression. Instead, we will use an iterative method called gradient descent.

1.6 Gradient descent

Definition: Gradient descent

Gradient descent is an optimization algorithm that iteratively updates parameters by moving in the opposite direction of the gradient of the loss function, in order to minimize the loss.

1.6.1 Loss function

We want parameters $\mathbf{w}, b$ that minimize loss averaged over m examples in our training set. We denote the average loss as $J(\mathbf{w}, b)$ with $\hat{y}_i = f(\mathbf{x}_i; \mathbf{w}, b)$ written as a function of inputs and parameters.

$$J(\mathbf{w}, b) = \frac{1}{m} \sum_{i=1}^m [y_i \log \sigma(\mathbf{w} \cdot \mathbf{x}_i + b) + (1 - y_i) \log [1 - \sigma(\mathbf{w} \cdot \mathbf{x}_i + b)]]$$

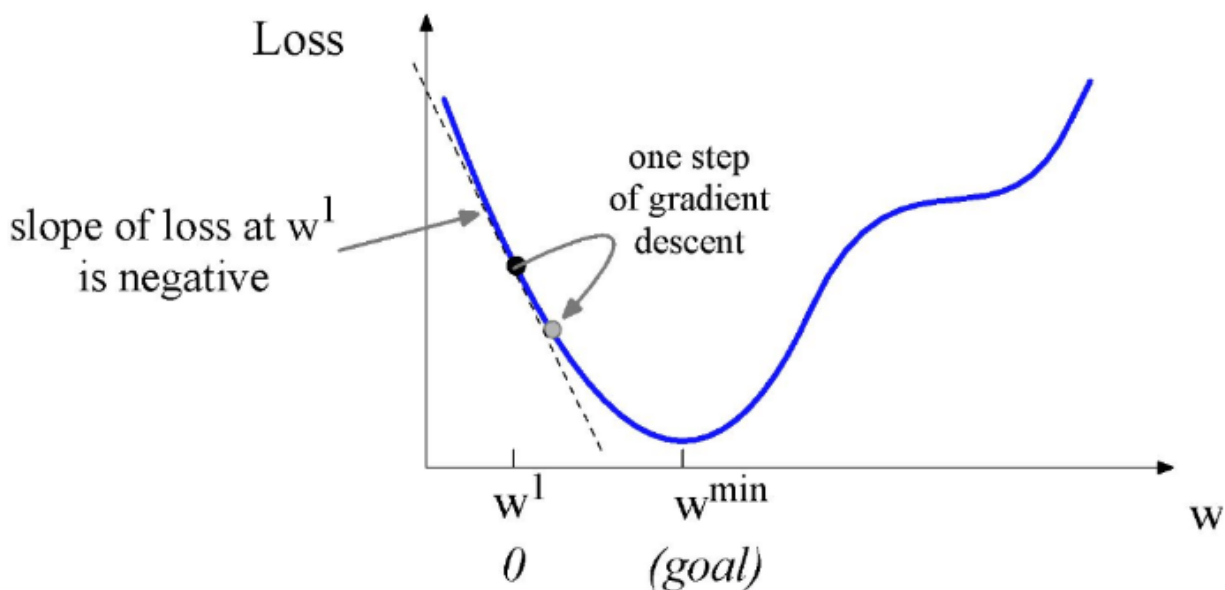
Now, we want to find:

$$\hat{\mathbf{w}}, \hat{b} = \underset{\mathbf{w}, b}{\operatorname{argmin}} J(\mathbf{w}, b)$$

1.6.2 Optimization

For logistic regression, the loss function is convex. Gradient descent from any point is guaranteed to find the minimum. Here is a visualization for a single scalar w : Given the current w , move in the reverse direction from the function slope.

L



The loss is a function of multiple parameters $w_1, \dots, w_n, b$:

$$L_{\text{CE}} = -[y \log \sigma(\mathbf{w} \cdot \mathbf{x} + b) + (1 - y) \log[1 - \sigma(\mathbf{w} \cdot \mathbf{x} + b)]]$$

The gradient of this loss with respect to the parameters θ will be a vector of partial derivatives.

$$\nabla_{\theta} L = \frac{\partial L}{\partial \theta} = \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \\ \vdots \\ \frac{\partial L}{\partial w_n} \\ \frac{\partial L}{\partial b} \end{bmatrix}$$

For models with n -dimensional weight vectors:

- For each dimension w_i , the gradient component i tells us the slope w.r.t that parameter.
- We express the slope as the partial derivative of L w.r.t w_i .
- The gradient is defined as the vector of these partial derivatives.

1.6.3 Learning rate

$$\mathbf{w}^{t+1} = \mathbf{w}^t - \eta \frac{d}{dw} L(f(\mathbf{x}; \mathbf{w}), y)$$

The gradient tells us:

- **Sign:** which way to move (left, right, up, down).
- **Magnitude:** how steep/big the update is.

To control the step size, the value of the gradient is weighted by a learning rate η . A higher value of η means $\mathbf{w}$ changes faster. This is considered a hyperparameter.

- If the learning rate is too high, it will take big steps and overshoot.
- If the learning rate is too low, it will take too long to converge.

1.6.4 Optimization (continued)

How do we find $\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \frac{1}{m} \sum_{i=1}^m L_{\text{CE}}(f(\mathbf{x}; \theta), y_i)$? We do the following iteratively:

- Compute the derivative of the loss function with respect to the parameters.
- Update the parameters to move in the opposite direction of the gradient.

The partial derivatives for logistic regression:

$$\boxed{\frac{\partial L}{\partial w_i} = [\sigma(\mathbf{w} \cdot \mathbf{x} + b) - y]x_i} \quad \boxed{\frac{\partial L}{\partial b} = \sigma(\mathbf{w} \cdot \mathbf{x} + b) - y}$$

1.6.5 Gradient descent formula

The final question for updating parameters θ based on the gradient is:

$$\theta^{t+1} = \theta^t - \eta \nabla_{\theta} L(f(\mathbf{x}; \theta), y)$$

Note

Gradient descent can be applied to any model with any number of parameters. Gradient descent is a general optimization method, not just for logistic regression!

1.7 Stochastic gradient descent

Definition: Stochastic gradient descent

Stochastic gradient descent (SGD) iteratively updates parameters. It approximates the true gradient by using a subset of data at each step.

At each step, a random subset of examples is sampled. Both the gradient is computed and the parameters are updated based on this subset.

$$\theta^{t+1} = \theta^t - \eta \frac{1}{B} \sum_{i=1}^B \nabla_{\theta} L(f(\mathbf{x}_i; \theta), y_i)$$

Batch / mini-batch: The subset of data used for SGD at a particular training step.

Batch size: The number of training examples in each batch. **This is a hyperparameter.**

1.7.1 Batching

In practice, it's computationally expensive to apply gradient descent to the entire training set simultaneously. This is due to the high computational/memory requirements for large datasets and the slow convergence. Instead, we employ the principle of batching.

Traverse the training set in batches, B examples at a time, updating parameters based on gradient w.r.t average loss of the batch.

$$J_B(\theta) = \frac{1}{B} \sum_{i=1}^B L(f(\mathbf{x}_i; \theta), y_i)$$

The gradient of the cross-entropy loss averaged over all B examples in a batch is the average of the individual sample gradients.

$$\nabla_{\theta} J_B(\theta) = \frac{1}{B} \sum_{i=1}^B \nabla_{\theta} L(f(\mathbf{x}_i; \theta), y_i)$$

1.7.2 Example

Assume initial parameters θ^0 are all zero: $w_1 = w_2 = b = 0$. let's perform a SGD update based on one example in the training set. Let $\mathbf{x} = [3, 2]$ with label $y = 1$. Then,

$$\begin{aligned} \nabla_{\theta} &= \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \\ \frac{\partial L}{\partial b} \end{bmatrix} = \begin{bmatrix} [\sigma(\mathbf{w} \cdot \mathbf{x} + b) - y]x_1 \\ [\sigma(\mathbf{w} \cdot \mathbf{x} + b) - y]x_2 \\ [\sigma(\mathbf{w} \cdot \mathbf{x} + b) - y] \end{bmatrix} \\ &= \begin{bmatrix} [\sigma(0) - 1](3) \\ [\sigma(0) - 1](2) \\ [\sigma(0) - 1] \end{bmatrix} \\ &= \begin{bmatrix} -1.5 \\ -1.0 \\ -0.5 \end{bmatrix} \end{aligned}$$

Now that we have the gradient vector, we compute the new parameter values with gradient descent. Set the learning rate $\eta = 0.1$.

$$\begin{aligned} \theta^1 &= \theta^0 - \eta \nabla_{\theta} L \\ &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} - (0.1) \begin{bmatrix} -1.5 \\ -1.0 \\ -0.5 \end{bmatrix} \\ &= \begin{bmatrix} -0.15 \\ -0.10 \\ -0.05 \end{bmatrix} \end{aligned}$$

1.7.3 Vectorization

Construct matrices and vectors that allow us to process a batch as a matrix/vector computation. Let the rows of matrix X represent input features of individual trainign samples with B examples in the batch and n input features:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{B1} & x_{B2} & \cdots & x_{Bn} \end{bmatrix}_{B \times n}$$

The vector $\mathbf{w}$ is the weight parameters. Let $\mathbf{b}$ be a B -dimensional vector repeating bias parameter b .

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}_{n \times 1} \quad \mathbf{b} = \begin{bmatrix} b \\ b \\ \vdots \\ b \end{bmatrix}_{B \times 1}$$

Then we can compute the weighted sums $\mathbf{w} \cdot \mathbf{x}_i + b$ simultaneously for all B examples with matrix multiplication.

$$X\mathbf{w} + \mathbf{b} = \begin{bmatrix} w_1x_{11} + w_2x_{12} + \cdots + w_nx_{1n} + b \\ w_1x_{21} + w_2x_{22} + \cdots + w_nx_{2n} + b \\ \vdots \\ w_1x_{B1} + w_2x_{B2} + \cdots + w_nx_{Bn} + b \end{bmatrix}_{B \times 1}$$

If we apply the sigmoid function to each element of this vector:

$$\hat{\mathbf{y}} = \sigma(X\mathbf{w} + \mathbf{b}) = \begin{bmatrix} \sigma(w_1x_{11} + w_2x_{12} + \cdots + w_nx_{1n} + b) \\ \sigma(w_1x_{21} + w_2x_{22} + \cdots + w_nx_{2n} + b) \\ \vdots \\ \sigma(w_1x_{B1} + w_2x_{B2} + \cdots + w_nx_{Bn} + b) \end{bmatrix}_{B \times 1} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_B \end{bmatrix}$$

1.8 Multinomial logistic regression

1.8.1 One-vs-rest classification

How can we generalize logistic regression to multi-class regression? **One-vs-rest classification.**

- Train K binary logistic regression models.
- Each classifier is trained to classify one class against all other classes.
- To make a prediction for a data point:
 - All K classifiers are used.
 - Each gives us a probability $P(y = 1|\mathbf{x})$ of its target class.
 - Take the class with the highest probability as the prediction.

For each class $c \in \{1, \dots, K\}$, we have a logistic regression model with its own weight vector $\mathbf{w}_c$ and bias term b_c . It is trained to maximize $P(y = 1|\mathbf{x}) = \sigma(\mathbf{w}_c \cdot \mathbf{x} + b_c)$ for class c .

Note

Because the sigmoid function is monotonically increasing, we don't need the probabilities. The weighted sums are enough!

Higher $\mathbf{w}_c \cdot \mathbf{x} + b_c \Rightarrow$ Higher $\sigma(\mathbf{w}_c \cdot \mathbf{x} + b_c)$.

To make a final decision, we compute all linear scores:

$$s_c(\mathbf{x}) = \mathbf{w}_c \cdot \mathbf{x} + b_c \quad \text{for each class } c \in \{1, \dots, K\}$$

and then choose the predicted class as the one with the largest score:

$$\hat{y} = \operatorname{argmax}_{c \in \{1, \dots, K\}} s_c(\mathbf{x})$$

where $\hat{y}$ is the predicted class.

Disadvantages

We need to train K different model for one task. In return, we get K unrelated probabilities that don't sum to 1.

1.8.2 Multinomial logistic regression

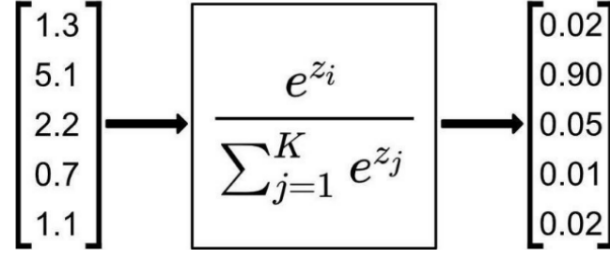
For multi-class classification, we have $y \in \{1, \dots, K\}$:

$$\begin{bmatrix} P(y=1|\mathbf{x}) \\ P(y=2|\mathbf{x}) \\ \vdots \\ P(y=K|\mathbf{x}) \end{bmatrix}$$

So instead of a single probability for the positive class, we need to output probabilities for each class: $P(y=1|\mathbf{x}), \dots, P(y=K|\mathbf{x})$ with

$$\sum_{k=1}^K P(y=k|\mathbf{x}) = 1$$

Instead of the sigmoid function, we can use the **softmax function** to squeeze K arbitrary values into probabilities summing to 1.



For input $\mathbf{x}$, the score (also called **logit**) for class c is $z_c = \mathbf{w}_c \cdot \mathbf{x} + b_c$. But probabilities need to be positive, so take the exponential e^{z_c} . But probabilities also need to sum to 1, so normalize.

The softmax function:

$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_{k=1}^K \exp(z_k)} \quad \text{for } 1 \leq i \leq K$$

This gives the softmax regression model:

$$f(\mathbf{x}; W, b) = \begin{bmatrix} P(y=1|\mathbf{x}; W, b) \\ P(y=2|\mathbf{x}; W, b) \\ \vdots \\ P(y=K|\mathbf{x}; W, b) \end{bmatrix} = \frac{1}{\sum_{k=1}^K \exp(\mathbf{w}_k^\top \cdot \mathbf{x} + b_k)} \begin{bmatrix} \exp(\mathbf{w}_1^\top \cdot \mathbf{x} + b_1) \\ \exp(\mathbf{w}_2^\top \cdot \mathbf{x} + b_2) \\ \vdots \\ \exp(\mathbf{w}_K^\top \cdot \mathbf{x} + b_K) \end{bmatrix} = \text{softmax}(W\mathbf{x} + \mathbf{b})$$

The model parameters are W and $\mathbf{b}$. W is a matrix with each row being the weight vector for a class. $\mathbf{b}$ is a vector of bias terms for each class.

$$W = \begin{bmatrix} \mathbf{w}_1^\top \\ \mathbf{w}_2^\top \\ \vdots \\ \mathbf{w}_K^\top \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_K \end{bmatrix}$$

Together, $\mathbf{w}_k$ and b_k define the linear decision boundary for class k .

The cross-entropy loss is also generalized to multiple classes. Now, $\hat{y}$ is a vector of K probabilities. Let y be a vector of targets, setting $y_c = 1$ for the true class and $y_c = 0$ for all other classes.

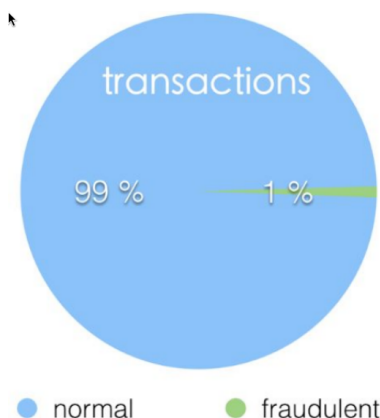
$$L_{CE} = - \sum_{k=1}^K y_k \log \hat{y}_k = - \sum_{k=1}^K y_k \log P(y = k|\mathbf{x})$$

This loss corresponds to what is called the negative log-likelihood. Without the minus sign, this is once again what we want to maximize with maximum likelihood estimation. For multinomial logistic regression, the cross-entropy loss is a function of all parameters $W, \mathbf{b}$.

$$\begin{aligned} L_{CE} &= - \sum_{k=1}^K y_k \log \hat{y}_k \\ &= - \log \hat{y}_c \quad (\text{where } c \text{ is the correct class}) \\ &= - \log \hat{P}(y_c = 1|\mathbf{x}) \\ &= - \log \frac{\exp(\mathbf{w}_c \cdot \mathbf{x} + b_c)}{\sum_{k=1}^K \exp(\mathbf{w}_k \cdot \mathbf{x} + b_k)} \end{aligned}$$

1.8.3 Imbalanced classes

For binary classification, the problem is framed as mapping inputs x to positive or negative labels. In a **balanced dataset**, both labels appear in **roughly equal numbers**. In an **imbalanced dataset**, **one label occurs much more often** than the other. For example, the training set of transactions labelled fraudulent vs. legitimate can have much higher proportion of legitimate transactions.



The problem is, does your dataset contain enough minority class examples to train a model properly?

- For mild imbalance (20%-40% minority) and sometimes for moderate imbalance (1%-20%), imbalance isn't a problem.
- For extreme imbalance (<1% minority), the model trains almost exclusively on one label and can't learn enough about the other label.

1.8.4 Handling imbalanced datasets

Downsampling majority class:

Reduce the number of majority examples by randomly selecting a subset of all majority examples to train on. For example, 1:10 $\rightarrow$ downsample by 1/5 times $\rightarrow$ 1:2. You can perform the sampling before training starts or at the start of every epoch.

Upsampling minority class:

Keep all majority class examples but duplicate minority class to match majority class size. For example, 1:10 $\rightarrow$ upsample by 10 times $\rightarrow$ 1:1.

Warning

This can lead to an **increase in the risk of overfitting** since the same minority examples are repeated multiple times!

Upsampling/downsampling is something you have to...

1. **Apply carefully:** it can improve performance, but it is not guaranteed to. It has trade-offs e.g. overfitting with upsampling and data loss with downsampling.
2. **Experiment with sampling ratios:** try different up/downsampling factors, e.g. 1:2, 1:3, 1:10, rather than assuming a perfectly balanced dataset is always best.
3. **Validate on a held-out validation set:** check that performance gains generalize beyond the resampled training data.

1.9 Text classification

Given a document d and a fixed set of classes $C = \{c_1, c_2, \dots, c_K\}$, can we accurately predict the true class $c \in C$?

Rule-based methods:

These are hard-coded rules based on combinations of words or other related features. The accuracy can be high if the rules are carefully refined by the expert(s), but building and maintaining these rules is expensive.

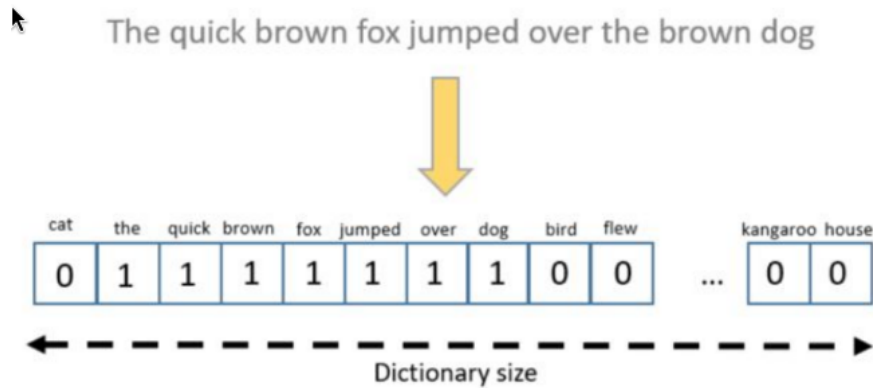
Machine learning methods:

For machines, given data D where inputs are documents and the outputs are classes, it can effectively learn a classifier $h : X \rightarrow Y$.

1.9.1 Bag-of-words

In a bag-of-words approach, we represent documents with word counts ignoring order.

1. Tokenize text into words and assign an integer ID to each unique word in the training dataset, called the vocabulary.
2. Count the word occurrences in each document.
3. Each document becomes a vector of indexed counts.



Word counts can be informative features for classification tasks. In logistic regression, each feature (word) will have a weight parameter w_i , which encodes its influence on predictions.

vocab = [a, boring, great, is, love, terrible, the, ...]
 $\mathbf{w} = [0.0, -2.2, +2.5, 0.0, +2.0, -2.5, 0.0, \dots], b = 0$

Positive review: "the movie is great and I love it"

- Feature vector = [0, 0, 1, 1, 1, 0, 1, ...]
- $\mathbf{w} \cdot \mathbf{x} + b = 1 \times (+2.5) + 1 \times (+2.0) = +4.5$
- $\sigma(4.5) \approx 0.99 \Rightarrow$ predict positive

Negative review: "the movie is boring and terrible"

- Feature vector = [0, 1, 0, 1, 0, 1, 1, ...]
- $\mathbf{w} \cdot \mathbf{x} + b = 1 \times (-2.2) + 1 \times (-2.5) = -4.7$
- $\sigma(-4.7) \approx 0.009 \Rightarrow$ predict negative

Most words never appear in a given document.

- Vocabulary can be very large (e.g. 100k words).
- Each document uses only a small fraction (100-1000 words), so most vectors are mostly zero (>99%).
- It is common to use sparse data structures for storage and efficiency.

Design choices affect what ends up in your vocabulary.

- **Text preprocessing/cleaning:** lowercase words, remove punctuation or number if not useful.
- **Tokenization:** the simple, straightforward approach is to split on whitespace characters. There are more sophisticated tokenizers available.
- **Vocabulary size control:** (based on training data frequencies!)
 - Keep only top N frequent words (tune N).
 - Don't include rare words (occurs less than some frequency threshold).

Tradeoff

A bigger vocabulary could produce better features, but might result in slower training!

1.9.2 Binary presence

Another option is to simplify bag-of-words to presence/absence of words.

- Feature = 1 if word appears, 0 otherwise.
- Ignores frequency of word within document.
- Useful when word presence is more informative than counts.

1.9.3 TF-IDF

The problem is that raw words counts are misleading because common words like "the" or "is" appear everywhere but add zero meaning. Truly informative words (e.g "remarkable", "special", "uninspiring") are rare and give a document its unique identity. We can use TF-IDF (term frequency - inverse document frequency) to scale down frequent, generic words and scale up rare, highly descriptive words to surface the most meaningful features.

TF (term frequency): number of times term t appears in document d divided by total number of terms in document d .

IDF (inverse document frequency):

- $\text{idf}(t) = \log(N/\text{df}(t))$, where
- N is the number of documents in the training set, $\text{df}(t)$ is the number of documents containing term t .
- $\text{idf}(t)$ is low if term t is common and high when term t is rare.

TF-IDF: $\text{tf} \times \text{idf}$

Extra

Usually the TF-IDF vector is L2-normalized. Reminder on L2-normalization:

$$x_{ij}^{\text{new}} = \frac{x_{ij}}{\sqrt{\sum_{i=1} x_{ij}^2}}$$

1.9.4 Example 1

Suppose we have 100 documents of 200 words each. The term "the" appears 20 times in document d and in 90 documents overall.

- $\text{tf}(\text{"the"}, d) = 20/200 = 0.1$
- $\text{idf}(\text{"the"}) = \log(100/90) = 0.046$
- $\text{tf-idf}(\text{"the"}, d) = 0.1 \times 0.046 = 0.0046$

1.9.5 Example 2

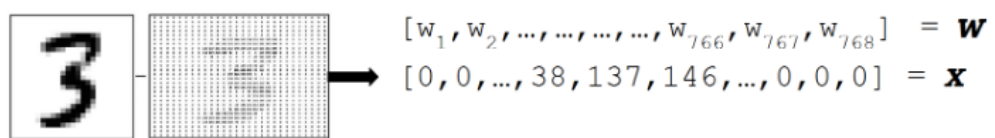
Suppose we have 100 documents of 200 words each. The term "happy" appears 5 times in document d and in 5 documents overall.

- $\text{tf}(\text{"happy"}, d) = 5/200 = 0.025$
- $\text{idf}(\text{"happy"}) = \log(100/5) \approx 3$
- $\text{tf-idf}(\text{"happy"}, d) = 0.025 \times 3 = 0.075$

1.10 Regularization

1.10.1 Interpreting logistic regression

After training, learned weights capture how each feature contributes to model predictions, giving us insight into what the model has learned.

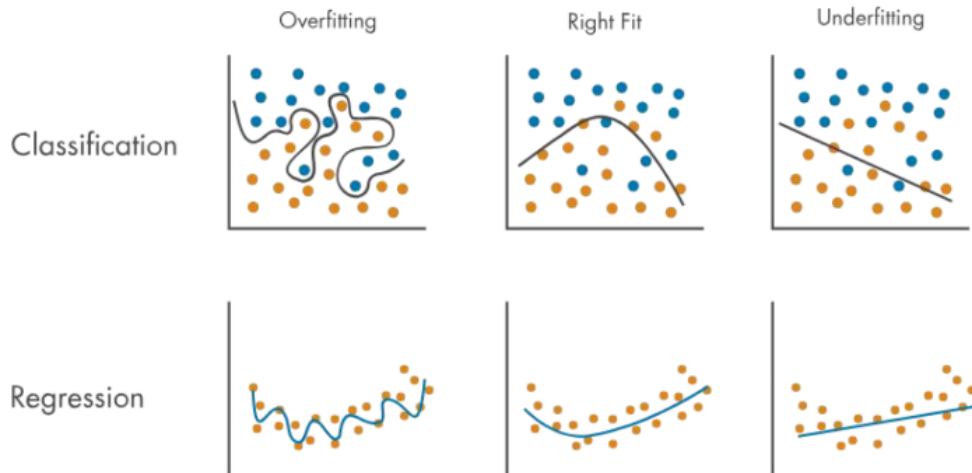


The magnitude of a weight $|w_i|$ reflects how strongly the associated feature influences the final predictions. Larger values have a stronger effect. The sign of w_i shows the direction of influence. Positive weights push the prediction towards class 1, while negative weights push the prediction towards class 0.

1.10.2 Overfitting

Definition: Overfitting

Overfitting is when your model is fitting the details of the training data so closely that the model doesn't generalize well to the test set. This happens when the model is over-parameterized. In other words, it has more parameters than required to model the data. The model is "too powerful", so it memorizes the training data. It predicts training data perfectly, but cannot generalize to new data.



How can overfitting happen for logistic regression?

- The weight w_i is learned such that $w_i x_i$ contributes to improved accuracy.
- If a feature x_i occurs only in one class in the training data, the model could learn very high weight w_i and perfectly predict y .
- The model might be over-relying on the presence of this feature, due to a peculiar property of the training data.
- The model will fail to generalize to a test set without this feature.

Tracking performance

High training accuracy, but low validation accuracy OR low training loss, but high validation loss.

- That's why it's important to track validation performance during training!
- If you only track training performance, this tells you nothing about how your model will generalize.

Potential causes of overfitting:

- **Wrong model for task:** model complexity exceeds available data.
- **Training data too small:** model will memorize rather than generalize.
- **Low-quality training data:** too much noise, which the model learns as if it is signal noise.
- **Too many training epochs:** eventually model fits training set too closely.

1.10.3 Early stopping

Definition: Early stopping

Stop training when the performance on held-out dataset stops improving to avoid overfitting.

Validation performance usually improve before degrading with overfitting. After this, improving training data fit comes at expense of generalization. Continually monitor performance on validation dataset and stop training where validation accuracy peaks.

1.10.4 Regularization

Add a regularization term $R(\theta)$ to the objective function.

$$\hat{\theta} = \operatorname{argmax}_{\theta} \left[\sum_{i=1}^B \log P(y_i | \mathbf{x}_i; \theta) - \alpha R(\theta) \right]$$

Choose $R(\theta)$ to penalize large parameters θ . Loss favors predicting data less well with small θ than predicting data well with big θ . More formally, we force optimization to balance maximizing the log-likelihood while minimizing weight magnitudes.

1.10.5 L2 Regularization

The regularization term is the sum of the squares of the weights. The L2 norm $\|\theta\|_2$ is the Euclidean distance of θ to the origin.

$$R(\theta) = \|\theta\|_2^2 = \sum_{i=1}^n \theta_i^2$$

The L2 regularized log-likelihood:

$$\hat{\theta} = \operatorname{argmax}_{\theta} \left[\sum_{i=1}^B \log P(y_i | \mathbf{x}_i; \theta) - \alpha \sum_{i=1}^n \theta_i^2 \right]$$

This is equivalent to minimizing the cross-entropy loss plus the L2 norm.

$$\hat{\theta} = \operatorname{argmax}_{\theta} \left[\frac{1}{B} \sum_{i=1}^B L_{\text{CE}}(\hat{y}_i, y_i) + \alpha \sum_{i=1}^n \theta_i^2 \right]$$

1.10.6 L1 Regularization

The regularization term is the sum of the absolute values of the weights. The L1 norm $\|\theta\|_1$ is the Euclidean distance of θ to the origin.

$$R(\theta) = \|\theta\|_1 = \sum_{i=1}^n |\theta_i|$$

The L1 regularized log-likelihood:

$$\hat{\theta} = \operatorname{argmax}_{\theta} \left[\sum_{i=1}^B \log P(y_i | \mathbf{x}_i; \theta) - \alpha \sum_{i=1}^n |\theta_i| \right]$$

This is equivalent to minimizing the cross-entropy loss plus the L1 norm.

$$\hat{\theta} = \operatorname{argmax}_{\theta} \left[\frac{1}{B} \sum_{i=1}^B L_{\text{CE}}(\hat{y}_i, y_i) + \alpha \sum_{i=1}^n |\theta_i| \right]$$

1.10.7 Regularization (continued)

L2 regularization produces weight vectors with many small weights:

$$[0.05, -0.01, 0.01, -0.005, 0.1, \dots]$$

L1 regularization tends to produce sparse solutions with some larger weights but many more weights set to zero.

$$[1.1, 0.0, -2.5, 0.0, 0.0, \dots]$$

$$\hat{\theta} = \operatorname{argmax}_{\theta} \left[\sum_{i=1}^B \log P(y_i | \mathbf{x}_i; \theta) - \alpha R(\theta) \right]$$

- α is a hyperparameter that controls the strength of regularization.
- During training, the model is biased towards smaller parameter values, which prevent over-reliance on certain features.
- These features could be unique to the training data, so not relying on them can reduce overfitting.